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Batch 15

Algorithm

1. Start.
2. Read a positive integer `n`.
3. Pass the address of `n` to `pivot_integer(&n)`.
4. In `pivot_integer()`:
   - Set `x` from `1` to `n`.
   - For each `x`, calculate `leftSum` as the sum from `1` to `x`.
   - Calculate `rightSum` as the sum from `x` to `n`.
   - If `leftSum == rightSum`, display `x` and return.
5. If no value of `x` satisfies the condition, display `-1`.
6. Stop.

Test Case Table

Test	Input n	Expected Output	Reason
1	8	`Pivot integer: 6`	`1+2+3+4+5+6 = 6+7+8 = 21`
2	1	`Pivot integer: 1`	`1 = 1`
3	4	`-1`	No pivot integer exists
4	49	`Pivot integer: 35`	Both sums are equal at `x = 35`
5	0	`Invalid input`	Not a positive integer

C program

```
#include <stdio.h>

void pivot_integer(int *n)
{
```

```
    int x, i;
    int leftSum, rightSum;

    for (x = 1; x <= *n; x++)
    {
        leftSum = 0;
        rightSum = 0;

        /* Sum from 1 to x */
        for (i = 1; i <= x; i++)
        {
            leftSum += i;
        }

        /* Sum from x to n */
        for (i = x; i <= *n; i++)
        {
            rightSum += i;
        }

        if (leftSum == rightSum)
        {
            printf("Pivot integer: %d\n", x);
            return;
        }
    }

    printf("-1\n");
}
```

Execution

